Steven Boada, Ph.D

Basic Information

(615) 200-0119
stevenboada@gmail.com
github.com/boada
linkedin.com/in/theboada

Professional Experience

Activision Publishing, Inc. — Boulder, Colorado

Senior Data Scientist, Gameplay — **January, 2024 - Present**

As part of the game science team, I work to make Call of Duty better through data. In this role, I:

- Partner with senior design leadership and cross-functional teams, including engineering and product management, to define key metrics, identify opportunities for game and business improvements, and drive data-informed decision making across the game.
- Lead initiatives to understand how gameplay systems (maps, modes, weapons, in-game communication) influence top-line KPIs and suggest design improvements based on data insights.
- Design, implement, and monitor experiments, A/B tests, and other methodologies to drive improvements in gameplay features and player engagement.
- Drive data quality and ensure instrumentation across the game to capture accurate and actionable data; collaborate closely with engineering teams to improve data pipelines and ensure seamless data flow.
- Lead the development and implementation of data models, production metrics, and visualizations to provide insights into player behavior, gameplay patterns, and overall game performance.
- Manage and mentor junior data scientists, offering guidance in experimental design, statistical analysis, and effective data visualization.

Senior Machine Learning Engineer — **January, 2021 - 2024**

- Designed, built, and maintained ML infrastructure and tools, including autoML, CI/CD (Jenkins, GitHub Actions), model tracking (MLflow), and orchestration (Airflow) to support data-driven decision-making.
- Crafted and implemented policy for external data ingestion and retention to ensure accessibility and proper data stewardship across the organization.
- Developed near-realtime applications (via Spark streaming) to identify and mitigate in-game cheating and malicious behaviors, achieving action times in the low 10s of seconds.
- Spearheaded the transition of ML infrastructure from AWS to GCP/GCS, optimizing scalability and performance.
- Created machine learning models focused on customer conversion, churn prediction, and behavioral segmentation using survival analysis, clustering, and tree-based techniques.
- Managed the day to day of a small team of junior data scientists to design, develop, and deploy recommendation systems that were integrated into Modern Warfare 2 (2022) and Modern Warfare 3 (2023).

Insight Data Science — New York, New York

Fellow — **January, 2020 - 2021**

- Helped optimize the way NYC health inspectors perform restaurant inspections in order to reduce the time critical health violations remain unaddressed.
- Trained a random forest in Python to prioritize NYC restaurant inspections based on environmental variables and their past inspection histories and provided the results to NYC through an API deployed on AWS.
- Resulted in NYC inspectors identifying $\sim$2.5% more violations in the first half of an inspection window, leading to critical violations being discovered up to 7 days earlier than by the current approach implemented by NYC.

Dept. of Physics and Astronomy, Rutgers University — New Brunswick, New Jersey

Postdoctoral Research Associate — **September, 2016 - 2021**

- Designed and built parallelized pipelines to process and analyze TBs of astronomical imaging; producing calibrated, standardized data catalogs and rigorous results leading to 2 peer reviewed publications and several hundred hours of telescope time.
- Contributed to open source, astronomy-focused, Python projects through bug fixes and feature additions: see photometrypipeline, astLib, and easyGalaxy on GitHub as examples.

Skills

Machine Learning: Linear Models, Decision Trees, SVM, Clustering, Deep Learning, Survival Analysis

CI/CD: Jenkins, Airflow, Docker
Software and Computing: Open Source Contributor, Python, DataBricks, MLFlow, SQL, AWS/GCP, Spark, and other cloud computing applications
Leadership: Experience organizing and leading workshops and collaboration meetings, Teaching and mentoring junior team members, Eagle Scout.

Education

Texas A&M University, College Station, Texas
- Ph.D, Physics (astronomy focus), 2016

The University of Tennessee, Knoxville, Tennessee
- M.S., Physics (astronomy focus), 2009
- B.S., Physics, 2007